DataWeave

Product Requirements Document

Next Version: v0.3.0

Native Windows-first PostgreSQL desktop workspace
Spreadsheet Data workspace | Visual Dataset Builder | SQL | Projects

Baseline: repository v0.2.9 (per release notes) | Document date: 24 October 2026 | Status: Draft for implementation

2. Document Metadata

Field	Value
Product	DataWeave, a native desktop workspace for PostgreSQL
Document	Product Requirements Document for the next version, v0.3.0
Baseline version	Repository v0.2.9 (Gradle 0.2.9, jpackage 1.2.9) as reported in docs/V0.2.9_RELEASE.md
Repository	github.com/nvsaigokul-sudo/DataWeave (public, branch main)
Primary platform	Windows 10 and 11 (x64). Linux and macOS installers exist; they are secondary and must not regress.
Document date	24 October 2026
Audience	Engineering team or coding agent implementing v0.3.0; product owner; tester
Requirement ID prefixes	FR (functional), UI (design and interaction), NFR (non-functional), TEST (verification), SEC (security), DM (data model)
Priority keys	M = must ship in v0.3.0; S = should ship, may slip with owner approval; C = could, deferred if time is short
Evidence labels	VERIFIED-DOC = stated in a repository document that was read. OWNER = stated by the project owner in the brief. ASSUMPTION = decision made here without evidence. UNVERIFIED = must be confirmed against source code.

Source reading performed

Two repository documents were read directly: README.md and docs/V0.2.9_RELEASE.md. The feature inventory, verification status, regression matrix, architecture document, build files and source tree were listed in the brief but were **not** read for this document. Every statement about code-level behavior is therefore labeled UNVERIFIED and is assigned to the Phase 1 audit (section 23).

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3. Executive Summary

DataWeave is an offline-first desktop application that connects to PostgreSQL, lets a user explore databases, inspect real table data in a spreadsheet-style grid, combine columns from several tables into a reusable dataset by dragging them onto a visual canvas, run SQL, organize the work in projects, and export results. It runs as a single process: Compose Desktop UI with an embedded Spring Boot layer, PostgreSQL JDBC with HikariCP, SQLite for local metadata, Flyway migrations, Caffeine caching, and a private Java 21 runtime packaged with jpackage.

Releases v0.2.4 to v0.2.9 spent most of their effort on connection and credential defects. The most recent release records that the Windows credential fix and recovery panel passed headless automated tests while the Windows GUI check is still **pending**. The owner now requires that the next version stop circling the connection screen and deliver the rest of the product with equal weight.

What v0.3.0 delivers

- **A professional design system** with crisp text, compact controls, dark and light themes, resizable panels and correct Windows display scaling.
- **A reliable connection and explorer foundation** with one shared connection state across all sections and explicit credential recovery.
- **A spreadsheet-style Data workspace** that opens any table as a virtualized grid with sort, filter, search, selection, copy, keyboard navigation and streamed export. View-only in v0.3.0.
- **A visual Dataset Builder** with drag-and-drop columns, foreign-key-aware joins (inner, left, right, full outer), aliases, filters, sorting, preview, and a saved definition stored independently of generated SQL.
- **An integrated SQL workspace** that shares the active connection, runs in a read-only transaction, shows duration, history, errors and supports cancellation.
- **Projects** that reference datasets, queries and export settings without copying PostgreSQL data.
- **A control inventory** where every visible button has a contract, and a test plan that separates automated runs from manual Windows GUI passes.

Key decisions made in this document

#	Decision	Reason
D-01	Keep Kotlin, Compose Desktop, embedded Spring Boot, JDBC, HikariCP, SQLite, Flyway, Caffeine and jpackage.	The brief requires it; nothing read so far contradicts it. Re-evaluate only if the Phase 1 audit finds a blocking defect.
D-02	The Data grid is view-only in v0.3.0. Cell editing is deferred.	Safe editing needs primary-key detection, transactions, conflict handling and undo. Silent writes are forbidden by the brief.
D-03	SQL workspace stays read-only (SELECT, WITH, EXPLAIN, SHOW, VALUES) and is enforced by a read-only database transaction, not only by text checks.	README states the editor is read-only SELECT today. Text filtering alone is bypassable.
D-04	Dataset joins are PostgreSQL-to-PostgreSQL within one database in v0.3.0. Cross-database and CSV-to-PostgreSQL joins are deferred.	A single query on one connection is verifiable and cancellable. README mentions CSV registration; its join behavior is UNVERIFIED.
D-05	Foreign-key relationships are auto-applied suggestions; name-similarity relationships are shown only as dismissible hints and never create a join without user confirmation.	Required by the brief.
D-06	Persist dataset definitions as versioned JSON in SQLite. Generated SQL is derived, cached and never the source of truth.	Lets users reopen and modify visually.

4. Product Vision

Vision statement. A user opens DataWeave, connects to a PostgreSQL server, finds a table, sees its real rows in a grid as comfortable as a spreadsheet, drags columns from several tables into one dataset, sees the joined result immediately, saves it, and exports it, without writing SQL unless they want to.

Product principles

Principle	Meaning for design and engineering
Real data only	No demo or placeholder data in any release build. Every visible number and row comes from the connected PostgreSQL server or local metadata.
Never lie about state	A saved profile is not a connection. A passed test is not an active session. A control that cannot work says so and why.
View is safe, edit is explicit	Nothing writes to PostgreSQL as a side effect of clicking, scrolling or pasting.
Beginner friendly, expert fast	Visual paths for every task, with keyboard shortcuts and SQL as the escape hatch.
Whole-product completeness	No release is accepted because one screen is polished while another is a stub.
Local first	No account, no cloud service and no telemetry is required. Secrets stay in the operating system credential store.
Density over decoration	Compact controls, sharp text, minimal effects, resizable panels.

Success measures

Targets below are proposals. Baselines are unknown and are measured in Phase 1 using the method in NFR-001. No benchmark results are claimed here.

Measure	Target for v0.3.0	How measured
Task completion without help: connect, open table, see rows	At least 9 of 10 first-time testers	Moderated Windows session, scripted task list (TEST-060)
Task completion: build a 3-table dataset and export CSV	At least 8 of 10 testers within 10 minutes	Moderated session (TEST-061)
Visible controls with a documented contract and a passing test	100 percent	Control inventory audit (FR-170)
Release defects reopened from earlier versions	Zero in the regression matrix	TEST-050 series

5. Existing Product and Technical Baseline

5.1 Technology stack (OWNER, partly VERIFIED-DOC)

Layer	Technology	Evidence
Language	Kotlin	OWNER
UI	Jetpack Compose Desktop	OWNER; README
Application services	Spring Boot embedded in the desktop process (single process, no separate server)	README
PostgreSQL access	PostgreSQL JDBC with HikariCP pooling	OWNER
Local metadata	SQLite stored under ~/.dataweave/ with Flyway migrations	README (path); OWNER (Flyway)
Caching	Caffeine	OWNER
Runtime and packaging	Java 21, jpackage installers with a private runtime; MSI and EXE on Windows, DEB and AppImage on Linux, DMG and PKG on macOS	README
Build and tests	Gradle; tasks documented: test, :desktop:runHeadlessSmoke, :desktop:run, :desktop:packageDistributionForCurrentOS	README
CI and release	GitHub Actions ci.yml and release.yml triggered by v* tags; SHA256 checksums published	README
Credential storage	OS keychain or credential manager; on Windows via Credential Manager using CRED_PERSIST_LOCAL_MACHINE since v0.2.9	README; V0.2.9 notes

5.2 Top-level sections

Home, Connections, Data, SQL, Datasets and Projects (OWNER). This PRD keeps those six and adds Settings as a dialog or footer entry, not a seventh primary section.

5.3 Capabilities documented in the README

- Save PostgreSQL connections with host, port, SSL and credentials in the OS credential store.
- Browse schemas and tables and view paginated table data.
- SQL editor limited to read-only SELECT with local query history.
- Dataset builder with column preview, CSV file registration and preview, and an inner-join preview.
- Streamed CSV export from table browse with an optional row limit.
- Local SQLite metadata; the README states no fake demo databases, no Docker and no browser UI.

5.4 Protected behavior

ID	Behavior that must not regress	Source
FR-001	GUI startup must call startForGui() and set headless(false). The earlier Windows HeadlessException must not return.	OWNER
FR-002	Credentials persist across application restart on Windows (CRED_PERSIST_LOCAL_MACHINE or an equivalent durable per-user mode).	V0.2.9 notes
FR-003	Connect and Test return a distinct missing_credential error kind, and the UI shows an inline credential recovery panel with host, database, user, masked password, Save credentials, Test, Connect and Open Connections.	V0.2.9 notes
FR-004	Edit connection from the explorer opens the correct saved profile in Connections.	V0.2.9 notes
FR-005	The Connections page scrolls so every field and button is reachable.	OWNER; v0.2.5 notes in README
FR-006	Connecting to PostgreSQL 18.x lists databases such as postgres, student, test and userdetails.	OWNER screenshot

6. Current-State Findings and Assumptions

6.1 Findings

ID	Finding	Evidence	Consequence
F-01	README names v0.2.5 as the latest published release while the release notes document v0.2.9.	README vs V0.2.9 notes	Documentation drift. Phase 1 reconciles README, releases and installers. Which installer is current is UNVERIFIED.
F-02	V0.2.9 reports automated tests passing but the Windows GUI manual check as PENDING.	V0.2.9 notes	No v0.2.9 UI behavior may be treated as verified. TEST-040 series must be executed.
F-03	Credential loss was caused by session-persistence mode, and recovery was a dead end inside the explorer.	V0.2.9 notes	Credential lifecycle is a regression-tested feature (TEST-010 to TEST-016).
F-04	Past failures included a connection reporting success while the explorer stayed on a loading state.	OWNER	Connection state and metadata load must be separate, visible states with timeouts (FR-020, FR-110).
F-05	README says SQL is read-only SELECT with local history, and the dataset builder is column preview, CSV registration and an inner-join preview.	README	Dataset Builder and SQL need substantial new work; they are not complete products today.
F-06	Table data is described as paginated; sort, filter, search, selection, copy and keyboard navigation are not mentioned.	README	Grid capabilities in section 13 are treated as new until the audit proves otherwise.
F-07	The visual design is described by the owner as basic, blurry, with oversized controls and empty panels.	OWNER	Design system work precedes feature screens (Phase 2). Blur may come from Compose scaling settings or bitmap icons; cause is UNVERIFIED.
F-08	Some visible controls did not perform their expected actions.	OWNER	Control inventory and contracts are mandatory (section 18).

6.2 Assumptions

ID	Assumption	Risk if wrong	Verification step
A-01	Source layout separates UI, services and persistence well enough to add features without a rewrite.	Phase 3 and 4 estimates grow.	Phase 1 module and dependency review
A-02	The current grid is a basic paginated table, not a virtualized grid.	Less new work than planned.	Phase 1 code read of the Data screen
A-03	SQLite schema has tables for profiles, history and datasets under Flyway control.	Migration plan changes.	Phase 1 read of Flyway scripts
A-04	Compose Desktop (Skia) can deliver a virtualized grid of 50,000 visible-window rows with smooth scrolling using LazyColumn or a custom canvas.	Custom canvas grid becomes necessary.	Phase 4 spike, 3 days, measured by NFR-002
A-05	Inter and JetBrains Mono (both SIL OFL) may be bundled.	Fall back to Segoe UI and Consolas.	License check in Phase 2
A-06	Test PostgreSQL instances can be started in CI with Testcontainers or an equivalent service container.	Integration tests run only on a developer machine.	Phase 1 CI check
A-07	Blur on Windows is caused by DPI handling or non-vector assets rather than by Compose itself.	Rendering fix needs JVM or Skia flags.	Phase 2 test at 100, 125, 150 and 200 percent scaling

6.3 Unresolved repository facts

- Whether FEATURE_INVENTORY.md, VERIFICATION_STATUS.md and REGRESSION_MATRIX.md exist and what they claim.
- Actual SQLite tables, Flyway versions, and whether existing profiles and history survive the planned migrations.
- Whether the explorer lazily loads metadata and how it handles roles that lack catalog access.

- How the existing inner-join preview is implemented and whether it joins PostgreSQL tables, CSV files, or both.
- Which UI strings and controls exist today; the Phase 1 audit produces the starting control inventory.

7. User Personas and Workflows

Persona	Profile	Needs	Success looks like
Priya, student analyst	Learning SQL; has a local PostgreSQL with sample databases	See tables as rows; combine two tables without writing JOIN	Builds a joined dataset by dragging and exports CSV in minutes
Marcus, backend developer	Writes SQL daily; checks data while building features	Fast explorer, keyboard navigation, SQL with history, cancel long queries	Never leaves the app to inspect a table or run a check
Elena, business analyst	Works from saved datasets for weekly reports	Reopen, tweak and re-export a dataset reliably	Opens a project, refreshes, exports within one minute
Raj, database administrator	Needs safe read access to production-like databases	Confidence nothing is written; clear SSL and timeout behavior	Trusts that DataWeave cannot modify data by accident

Primary workflows

ID	Workflow	Steps and requirement links
WF-01	First connection	Home shows Connect action, then new profile form (FR-100), Test (FR-103), Save (FR-102), Connect (FR-104), explorer lists databases (FR-110).
WF-02	Return visit	Home lists saved profiles with status; select profile; if credential is missing the recovery panel appears inline (FR-003, FR-106); connect.
WF-03	Inspect a table	Expand database, schema, select table; Data workspace opens grid with first page (FR-120 to FR-140); sort, filter, search, copy; export.
WF-04	Build a dataset	Open Datasets, New dataset; drag columns from two or more tables; accept FK join; set aliases and filter; preview; save (FR-150 to FR-171).
WF-05	Run SQL	Open SQL with the same connection and database; run SELECT; read results; check history; export (FR-180 to FR-195).
WF-06	Organize work	Create a project; attach datasets and queries; reopen later; handle missing source (FR-200 to FR-215).
WF-07	Recover from failure	Server unreachable, credential missing, permission denied, query timeout: each shows cause, safe next action and Retry (FR-300 series).

These workflows share one **Workspace Context** (active connection, database, schema, selected table). Selecting a table in the explorer updates the context; the Data workspace, the Dataset Builder source list, and the SQL editor schema hints all read the same context. The context is held in one state holder and exposed to every section (FR-021).

8. Scope and Exclusions

8.1 In scope for v0.3.0

Area	Included
Design	Design system, dark and light themes, theme persistence, high-DPI correctness, resizable panels, accessibility baseline
Connections	Profile CRUD, SSL modes, Test, Connect, Disconnect, Reconnect, credential recovery, consistent status
Explorer	Connection to databases to schemas to tables and views to columns, keys and indexes; lazy load, filter, refresh, retry, context menus, cache
Data	Virtualized spreadsheet grid, sort, filter, search, selection, copy, type-aware formatting, export current result, view-only
Datasets	Visual builder with drag-and-drop, FK joins, four join types, aliases, filters, sort, preview, save, reopen, export
SQL	Editor, run, cancel, status, duration, result grid, errors, history, export; read-only
Projects	Create, open, save, rename, duplicate, delete; reference datasets and queries and export settings
Quality	Control inventory, automated tests, Windows manual test scripts, upgrade test, packaged installer verification

8.2 Explicitly out of scope

- Inline editing, insert and delete of PostgreSQL rows (deferred to v0.4.0, preconditions in FR-141).
- SQL writes, DDL and schema management from the UI (the README description mentions schema management and data cleaning as long-term direction; neither is part of v0.3.0).
- Cross-database joins, joins between CSV files and PostgreSQL tables, other database engines.
- Cloud sync, accounts, telemetry, collaboration, AI features.
- Advanced SQL editor features: autocomplete, formatting and full syntax highlighting are not promised. Basic keyword highlighting is a Could (UI-090).
- ER diagram reverse engineering and visual schema design.
- Code signing and notarization changes (documented per release; unchanged).

8.3 Platform stance

Windows 10 and 11 are the verified platform for v0.3.0 acceptance. Linux and macOS packages continue to build in CI; their GUI behavior is tested only by automated tests and marked unverified in release notes unless a manual pass is done.

9. Information Architecture

9.1 Navigation model

A left navigation rail holds Home, Connections, Data, SQL, Datasets and Projects, with Settings and theme toggle at the bottom. A persistent **status bar** runs along the bottom of the window and shows connection name, server version, current database, and background activity. A persistent **explorer panel** is shown beside Data, SQL and Datasets and can be collapsed.

Section	Purpose	Explorer panel	Primary content
Home	Start here: recent connections, recent datasets and projects, quick actions	Hidden	Connect card, saved profiles, recent items
Connections	Manage profiles	Hidden	Profile list and editor form
Data	Inspect table rows	Shown	Grid with toolbar, tabs for open tables
SQL	Run queries	Shown	Editor and result grid, history drawer
Datasets	Build and manage datasets	Shown as source list	Canvas, column list, preview
Projects	Organize reusable work	Hidden	Project list and project detail

9.2 Cross-section behavior

- **Tabs.** Data and SQL support multiple tabs; a tab holds its own sort, filter and scroll state. Tab state is restored on restart when the connection is available (FR-142).
- **Deep links.** Right-click a table: Open data, Open in SQL (inserts a SELECT), Add to dataset, Copy qualified name.
- **Status consistency.** Connection state has exactly five values shown identically everywhere: Disconnected, Connecting, Connected, Reconnecting, Failed.
- **Minimum window.** 1024 by 640 logical pixels. Below 1100 pixels width, the explorer collapses to an icon rail.

10. Design System

The goal is a compact, sharp, professional tool. Values are in density-independent pixels (dp). All sizes below are proposals for implementation and are validated visually in Phase 2 at 100, 125, 150 and 200 percent Windows scaling.

10.1 Typography

Token	Font	Size / line	Use
type.display	Inter SemiBold	20 / 28	Page titles, empty-state headings
type.title	Inter SemiBold	14 / 20	Panel titles, dialog titles
type.body	Inter Regular	13 / 18	Default UI text, inputs, buttons
type.small	Inter Regular	12 / 16	Status bar, captions, helper text
type.grid	Inter Regular with tabular numerals	13 / 20	Grid cells
type.mono	JetBrains Mono	13 / 20	SQL editor, SQL previews, identifiers

Minimum text size anywhere is 12 dp. Fonts are bundled as resources; if loading fails the app falls back to Segoe UI and Consolas without error (UI-004).

10.2 Spacing, sizing and shape

Token	Value	Notes
Spacing grid	4 dp base: 4, 8, 12, 16, 24, 32	All padding and gaps use these steps
Control height	Compact 28, Default 32, Large 40 (primary dialog actions only)	No control taller than 40 dp
Icon size	16 dp in controls, 20 dp in navigation rail	Vector (SVG or ImageVector) only; no bitmap icons
Grid row height	24 dp default, 28 comfortable, 20 dense	User setting; persisted
Corner radius	4 dp controls, 6 dp panels and dialogs	No shadows except dialogs (one subtle elevation)
Border	1 dp hairline in border token	Pixel-aligned to avoid blur at fractional scales
Navigation rail	56 dp wide	Icon with label below at 12 dp
Explorer panel	Default 280 dp, min 200, max 480, resizable	Width persisted

10.3 Color tokens

Token	Dark	Light	Use
surface.base	#14171C	#F7F8FA	Window background
surface.panel	#1B1F26	#FFFFFF	Panels, grid background
surface.raised	#232832	#EEF1F5	Toolbars, headers, sticky header row
border.default	#2E3542	#D5DAE2	Dividers and outlines
text.primary	#E6EAF0	#1B2430	Body text (contrast at least 4.5:1)
text.secondary	#9AA6B6	#566170	Secondary text (at least 4.5:1)
accent	#4C8DFF	#1F6FEB	Primary buttons, selection, focus ring
null.text	#7A8798	#6B7685	NULL marker, italic
status.ok / warn / error	#3FB950 / #D29922 / #F85149	#1A7F37 / #9A6700 / #CF222E	Status, validation; never color alone, always with icon or text

10.4 Components and states

ID	Requirement	Pri
UI-001	Every interactive component defines default, hover, pressed, focused, disabled and loading states. Focus ring is 2 dp accent with 2 dp offset and is visible in both themes.	M
UI-002	Buttons: Primary (filled accent), Secondary (outlined), Tertiary (text), Destructive (red outline, filled on confirm dialog), Icon button (28 by 28 dp minimum hit target 28).	M
UI-003	Inputs have a visible label or accessible name, inline validation text below, and a clear button where useful. Password fields are masked with a reveal toggle that auto-hides on focus loss.	M
UI-004	Fonts and icons are bundled; missing resources fall back silently to system fonts and a text label.	M
UI-005	Loading uses a determinate or indeterminate 2 dp progress bar at the top of the affected panel; full-panel spinners are used only for first load and have a 10 s timeout leading to an error with Retry.	M
UI-006	Empty states are compact (maximum 240 dp tall), state the reason in one sentence, and offer one action. No oversized blank containers.	M
UI-007	Error banners are inline within the failing panel with: plain-language cause, technical detail expander, and Retry. Toast notifications are used only for transient success such as Copied.	M
UI-008	Dialogs are modal, keyboard dismissible with Escape (except while a destructive action is executing), trap focus, and place primary action at the right.	M
UI-009	Panels are separated by draggable splitters with a 6 dp hit area; double-click resets width; sizes persist per section.	M
UI-010	Tooltips appear after 600 ms on icon-only controls and show shortcut keys.	S
UI-011	No gaussian blur, translucency or decorative gradient anywhere. Animations are limited to 150 ms opacity or size changes and respect an Reduce motion setting.	M

10.5 Themes

- Three choices: Dark, Light, System. System follows Windows app mode at launch and when it changes.
- The choice is stored in the SQLite settings table and applied before the first frame to avoid a flash (UI-020).
- Both themes pass the contrast checks in NFR-040.

10.6 High-resolution rendering

- UI-030: Text and icons must be crisp at 100, 125, 150 and 200 percent Windows scaling and on a mixed-DPI multi-monitor setup. Acceptance is a screenshot review at each scale plus a zoomed pixel check of text edges and icon strokes.
- UI-031: Layout uses dp units only; no hard-coded pixel sizes. Borders align to device pixels.
- UI-032: If blur persists after bitmap assets are removed, the Phase 2 spike evaluates the JVM and Skiko DPI flags and records the chosen setting in ARCHITECTURE.md.

11. Screen-by-Screen Specifications

11.1 Home

Region	Content and behavior
Header	Application name, theme toggle, Settings.
Connect card	Always visible. Disconnected: lists saved profiles (name, host, database, last connected) with Connect on each row plus New connection. Connected: shows server version, current database, Disconnect and Open Data.
Credential recovery	If a selected profile lacks a stored password, an inline form (masked password, Save credentials, Test, Connect, Open Connections) replaces the Connect row for that profile (FR-106).
Recent	Recent datasets, queries and projects with unavailable-source badges. Empty state: one line with Create dataset.
Never shown	A bare filter field with the text Select or connect a profile and no action (earlier defect).

11.2 Connections

Region	Content and behavior
Profile list (left, 320 dp)	Search field, profiles with status dot, New profile button. Context menu: Connect, Edit, Duplicate, Delete.
Editor (right, scrollable)	Name, host, port, database, username, password, SSL mode, advanced (connect timeout, application name). Actions row sticky at bottom: Test, Save, Connect, Cancel. The form scrolls; the action row never scrolls out of view.
Result area	Test and connect results appear inline under the form with server version, latency and SSL status, or the error with next step.

11.3 Data

Region	Content and behavior
Explorer panel	Database tree (section 12).
Tab strip	One tab per opened table or view with schema-qualified title; close button; overflow menu.
Toolbar	Refresh, Columns chooser, Filter, Search, Row height, Export, row count and query status.
Grid	Virtualized grid (section 13) with sticky header and row numbers.
Footer	Row range shown, total (estimated or exact), page controls or loading indicator, last refresh time, elapsed query time.

11.4 SQL

Region	Content and behavior
Editor (top, resizable)	Monospace editor with line numbers, Run (Ctrl+Enter), Cancel, Clear, History.
Context chip	Shows active connection and database; changes with the explorer selection (FR-184).
Results (bottom)	Tabbed: Results grid, Messages. Footer: row count, duration, Export.
History drawer	Right drawer listing past queries with time, database, duration, status; click loads into editor; filter; delete one or clear all.

11.5 Datasets

Region	Content and behavior
Dataset list	Saved datasets with name, sources, last modified, status (Valid, Needs attention). New, Open, Duplicate, Delete.

Region	Content and behavior
Builder (section 14)	Left: source tables and columns. Center: canvas with source cards and join links. Right: dataset columns list with alias, order and filter. Bottom: preview grid and generated SQL (read-only view).

11.6 Projects

Region	Content and behavior
Project list	Name, description, item count, last opened, source connection.
Project detail	Metadata (name, description), items (datasets, queries, export presets) with availability status, Open, Remove, Add item.

11.7 Settings dialog

Theme, grid row height, default page size, default export options, query timeout default, history retention, log folder link, and About with version and build id. Every control persists on change (FR-250).

12. Database Explorer

The explorer is a lazy-loaded tree built on real PostgreSQL catalog queries. It shows only objects the connected role can discover and says so when a list is empty.

12.1 Hierarchy

Level	Node	Source	Notes
1	Connection	Profile and session	Shows status dot, server version, Disconnect
2	Database	pg_database where datallowconn	Listing needs no per-database session. Expanding a database opens a pooled session to that database on demand.
3	Schema	pg_namespace (hide pg_toast and temp by default; toggle to show system schemas)	Lazy
4	Tables, Views, Materialized views, Foreign tables (groups)	pg_class with relkind filter	Count badge after load
5	Columns, Keys, Indexes, Foreign keys	pg_attribute, pg_constraint, pg_index	Columns show type, nullability, PK or FK badge

12.2 Requirements

ID	Requirement	Priority
FR-110	On Connect, load the database list within the connection timeout and show Connected only after the list returns or fails. A failed list shows an error with Retry and keeps the session open.	M
FR-111	Each node loads children only on first expansion; a spinner appears on the node; a 15 s timeout converts the spinner to an error row with Retry.	M
FR-112	Selecting a table or view single-click highlights it and updates Workspace Context; double-click or Enter opens it in Data (FR-120).	M
FR-113	Filter field matches object names case-insensitively across loaded nodes and triggers lazy load for schemas on demand with a visible Searching state; clear restores the previous expansion.	M
FR-114	Refresh on any node invalidates its cached subtree and reloads; Refresh all invalidates the connection cache.	M
FR-115	Switching database via the tree or a status-bar picker changes Workspace Context and closes dependent tabs only after confirmation if they have unsaved dataset edits.	M
FR-116	Context menu per node: Table: Open data, Open in SQL, Add to dataset, Copy name, Copy qualified name, Refresh, Properties (columns, keys, indexes, estimated rows, size). Schema: Refresh, Copy name. Database: Switch to, Refresh.	M
FR-117	Empty states: no databases, no schemas, no tables, no permission. Each states the reason and does not look like loading.	M
FR-118	Metadata cache (Caffeine) is keyed by connection id, database and object path, expires after a configurable TTL (default 5 minutes), and is invalidated on Refresh, Disconnect and profile edit.	M
FR-119	Keyboard: Up and Down move, Right expands, Left collapses, Enter opens, F5 refreshes, Ctrl+F focuses filter.	M

12.3 States

State	Display	Next actions
Disconnected	Message plus Connect button and last profile	Connect, choose profile
Loading	Node spinner; progress bar on panel for first load	Cancel for loads over 3 s
Error	Row with message, detail expander	Retry, Edit connection
Empty	One-line reason	Refresh

State	Display	Next actions
Missing credential	Credential recovery panel (FR-003)	Save credentials, Connect

13. Spreadsheet Data Workspace

Opening a table shows real rows in a grid that behaves like a spreadsheet for reading. The workspace is **view-only**; its toolbar shows a persistent View mode label so users never wonder if a click can change data.

13.1 Grid features

ID	Requirement	Priority
FR-120	Opening a table runs a bounded query (default page size 500, setting 100 to 5000) and shows rows within the performance target NFR-002; the grid shows column headers and a first-column row number gutter.	M
FR-121	Header row and row-number gutter are sticky during vertical and horizontal scroll respectively. Headers show column name, type badge, PK or FK icon, sort indicator and filter indicator.	M
FR-122	Rows are virtualized: only rows in the viewport plus a buffer are composed. Memory holds at most a configurable window (default 5,000 rows) of fetched rows; older pages are discarded.	M
FR-123	Column resize by dragging header edge; double-click edge auto-fits to content of loaded rows (cap 600 dp); widths persist per table and connection. Reorder and hide via the Columns chooser.	M
FR-124	Sorting: click header cycles ascending, descending, none; Shift+click adds secondary keys up to 3. Sorting runs on the server using ORDER BY with quoted identifiers and appends the primary key as a tiebreaker when available so paging is stable.	M
FR-125	Filtering: per-column filter row or dialog with type-aware operators (text: equals, contains, starts with, ends with, is null; number and date: equals, between, greater, less, is null; boolean). Filters are combined with AND and executed on the server with bound parameters.	M
FR-126	Search: a single box searches text-castable visible columns with ILIKE on the server. Searches over very large tables are warned and can be cancelled (FR-136).	M
FR-127	Pagination: Previous, Next, page size, Go to row. For tables with a unique primary key and no custom sort, use keyset paging; otherwise use LIMIT and OFFSET and warn when offset exceeds 100,000.	M
FR-128	Row count: show an estimate from pg_class.reltuples immediately labeled approximate; offer Count exactly as an explicit, cancellable action. Filtered counts are exact only on request.	M
FR-129	Selection: click selects a cell; row-number click selects a row; header click selects a column; Shift extends; Ctrl toggles; Ctrl+A selects loaded rows. Selection summary (count, numeric sum and average) is shown in the footer.	M
FR-130	Keyboard navigation: arrows move; Home, End, Ctrl+Home, Ctrl+End; Page Up, Page Down; Tab moves right; F2 and Enter open the full-value viewer (not an editor).	M
FR-131	Copy: Ctrl+C copies selection as tab-separated text with optional headers (Ctrl+Shift+C). Copy as CSV, Copy as JSON and Copy column name are in the context menu. Column copy copies loaded values and states if the column is only partially loaded.	M
FR-132	NULL is shown as an italic NULL marker in null.text color, distinct from an empty string, which shows as an empty cell with an outlined placeholder on hover.	M
FR-133	Long text is truncated with an ellipsis on one line; hovering for 600 ms or pressing F2 opens the value viewer with wrap, copy, and for JSON a formatted view. Values over 1 MB are truncated at fetch with a clear Truncated badge and a Load full value action.	M
FR-134	Type-aware formatting: numbers right-aligned with tabular numerals and no silent rounding of numeric or decimal; dates and timestamps in ISO format with timezone indicator for timestamptz; booleans as checkmark plus text; bytea as hex preview with size; arrays and json as compact text; uuid monospace.	M
FR-135	Refresh re-runs the current query, preserving sort, filters, column widths and, where possible, scroll position; shows last refreshed time.	M
FR-136	Cancel: every running grid query has a Cancel control; cancellation issues a JDBC statement cancel and falls back to pg_cancel_backend; the UI returns to idle within 2 s.	M
FR-137	Export current result: CSV (UTF-8, optional BOM, delimiter, quote, header, NULL representation) streamed to disk in batches of 10,000 rows with progress, Cancel and a row limit option. The export re-runs the same query (sort and filters) and does not rely on loaded pages. Excel export is deferred.	M
FR-138	Status line shows query state (Idle, Loading, Done, Error, Cancelled), rows shown, elapsed milliseconds and the effective filter and sort summary.	M
FR-139	Show query: a read-only popover shows the generated SQL for the current view with a Copy button (identifiers quoted; filter values shown as parameters).	S

ID	Requirement	Priority
FR-140	Unsupported types render as text via cast with a type badge rather than failing the whole page.	M

13.2 View versus edit

ID	Requirement	Priority
FR-141	Inline editing is not available in v0.3.0. Cells show no editor on click. Any attempt to type with a cell focused does nothing and shows a status-bar hint 'Data workspace is read-only'. Future editing requires: unique key detection, explicit Edit mode toggle, staged changes with diff, single transaction commit, optimistic conflict check, rollback on error, and an audit entry. These are listed for v0.4.0 and are not promised.	M
FR-142	Open tabs, per-table sort, filters, hidden columns and widths persist across restart and restore when the connection and table still exist; otherwise the tab shows 'Source unavailable' with Retry and Close.	S

13.3 States

State	Behavior
Loading first page	Skeleton rows are not used; a top progress bar and the headers (from metadata) show immediately.
Empty table	Headers shown; body message: This table has no rows. Refresh.
No matches	Body message: No rows match the current filters, with Clear filters.
Error	Banner with cause (permission denied, relation does not exist, timeout, connection lost), Retry and Reconnect. Prior rows remain visible, dimmed, if any were loaded.
Connection lost	Grid is retained; status Reconnecting; Retry automatic once, then manual.

14. Visual Dataset Builder

The Dataset Builder lets a user compose a result by dragging columns from several tables. The saved artifact is a **dataset definition** (structured data), and SQL is generated from it on demand and validated before execution.

14.1 Layout and interaction

Area	Behavior
Source panel (left)	Tables from the explorer. Search, expand to see columns with type badges. Each column row is a drag source and has an Add button. Tables can be added to the canvas by dragging or via Add table.
Canvas (center)	Source cards for each table with their selected columns highlighted. Join links connect cards, labeled with join type and key pairs. Cards can be moved, collapsed and removed. Pan and zoom 50 to 200 percent.
Dataset columns (right)	Ordered list of output columns: source, column, alias, type, visible toggle, remove. Drag to reorder.
Preview (bottom)	First 200 rows of the real query result in the same grid component, plus a Generated SQL tab (read-only, copyable) and a Validation tab.
Toolbar	New, Save, Save as, Preview, Refresh preview, Filter, Sort, Export, Undo, Redo, Open in SQL.

14.2 Drag-and-drop contract

ID	Requirement	Pri
UI-040	Drag sources: column row in source panel, column row on a canvas card, row in the dataset column list (reorder). Drag preview is a chip with table.column and type.	M
UI-041	Valid drop targets: dataset column list (insert at indicated position), canvas empty area (adds the table and the column), a canvas card (adds the column from that same table). Valid targets show an accent outline and an insertion line.	M
UI-042	Invalid drops (a column dropped on a different source's key area, onto a read-only panel, or a duplicate of an already selected column without alias) show a not-allowed cursor and, on drop, an inline message explaining why. Nothing changes.	M
UI-043	Auto-scroll when dragging near the edge of a list or canvas; Escape cancels the drag.	M
UI-044	Keyboard alternative: select a column and press Space or Enter on Add to dataset; Alt+Up and Alt+Down reorder; Delete removes; context menu offers Add to dataset, Add table to canvas, Move up, Move down, Remove. Every pointer-only action has a keyboard path.	M
UI-045	Small windows (below 1100 dp wide): panels become tabs (Sources, Canvas, Columns, Preview), the canvas switches to a list-based join editor, and drag-and-drop remains for lists. Below 900 dp the builder is usable via keyboard and menus only.	M
UI-046	Undo and redo (at least 50 steps) cover add, remove, reorder, alias, join and filter changes.	S

14.3 Relationships and joins

ID	Requirement	Priority
FR-150	When a second table is added, the builder looks for foreign keys between the tables (both directions, including composite keys) from pg_constraint and proposes joins. FK-derived joins are labeled Foreign key and are applied as a LEFT JOIN by default after a visible one-click Accept, never silently.	M
FR-151	Columns with the same name or similar names but no foreign key are surfaced as Suggested (dashed link, lighter style, label 'Name match, not a constraint'). They never become a join until the user clicks Use this. The UI states they are unverified.	M
FR-152	Manual join: drag a column from one card onto a column of another card, or use Add join in the list editor. The user picks left column(s) and right column(s); composite keys are supported; multiple conditions combine with AND.	M
FR-153	Join types: INNER, LEFT, RIGHT, FULL OUTER. Each shows a plain-language description and a small Venn icon. Default for FK is LEFT from the first table; the user changes it per join.	M
FR-154	Type compatibility check: joining incompatible types shows a warning with the cast option; casts are explicit in the definition and visible in SQL.	M

ID	Requirement	Priority
FR-155	Every table after the first must be connected by at least one join; otherwise validation fails with 'Table X is not joined', preventing accidental cartesian products. A deliberate CROSS JOIN is not offered in v0.3.0.	M
FR-156	Row multiplication warning: when the join key on the right side is not covered by a primary key or unique index, show 'This join may multiply rows' with the reason.	S
FR-157	Self-joins are allowed; each table instance receives a distinct instance alias (customers, customers_2) editable by the user.	S

14.4 Columns, aliases, filter and sort

ID	Requirement	Priority
FR-160	Output column names must be unique. When two selected columns share a name, the builder auto-suggests table-prefixed aliases (orders_id, customers_id), highlights the conflict, and blocks Preview until resolved or accepted.	M
FR-161	Aliases are validated: 1 to 63 characters, no leading digit unless quoted; quoted identifiers are always used in SQL so reserved words are safe.	M
FR-162	Filters: visual condition builder with groups (AND, OR), type-aware operators and bound parameter values; filters reference source columns even if the column is not in the output.	M
FR-163	Sorting: up to 5 sort keys on any source or output column with direction and NULLS FIRST or LAST.	M
FR-164	Limit: optional row limit for preview and for export; preview always applies its own limit (default 200).	M
FR-165	Removing a column used in a filter, sort or join prompts with the dependent items and offers Remove all or Cancel.	M

14.5 Query generation and validation

ID	Requirement	Priority
FR-170	SQL is generated from the definition by a pure, unit-tested generator. Identifiers are quoted with doubled embedded quotes; values are bound as parameters, never concatenated.	M
FR-171	Before Preview or Export, validation checks: sources exist, columns exist and types match the recorded types, joins complete, aliases unique. On the server, validation executes EXPLAIN without running the query when cost is a concern, or runs with LIMIT 0 to confirm syntax and types.	M
FR-172	Preview runs in a read-only transaction with a statement timeout (default 30 s) and the preview limit; Cancel is available.	M
FR-173	Schema drift: when a saved definition references a missing table or column or a changed type, the dataset opens in Needs attention state, highlights the broken items, keeps all other work, and offers Remap column or Remove item. It never deletes the definition automatically.	M
FR-174	Save, Save as, Rename, Duplicate and Delete work from the builder and the Datasets list; Delete asks for confirmation and states whether projects reference the dataset.	M
FR-175	Export from a dataset streams the full query (not only the preview) to CSV with the same options as FR-137.	M
FR-176	Open in SQL copies the generated SQL into a new SQL tab; edits there do not change the dataset.	S

14.6 Saved dataset definition (logical content)

The stored JSON contains: schemaVersion, id, name, description, createdAt, modifiedAt, connectionRef (profile id and database name; no secrets), sources (instance alias, schema, table, relkind), columns (ordered; source alias, column, output alias, visible, recorded type), joins (type, left alias, right alias, condition pairs, origin: foreign_key, manual or suggested_accepted, optional cast), filters (tree), sorts, limits, and canvas layout (card positions and zoom). The generated SQL is not authoritative and may be cached with a hash of the definition (DM-020).

15. SQL Workspace

The SQL workspace uses the same active connection, database and Workspace Context as Data and Datasets. v0.3.0 remains read-only, consistent with the README description. Read-only is enforced by the database session, not only by text inspection.

ID	Requirement	Priority
FR-180	Editor: multi-line monospace text area with line and column indicator, undo and redo, tab indentation, Ctrl+Enter to run the selection or the statement at the cursor, F5 to run all, and basic bracket and quote matching. Syntax highlighting, autocomplete and formatting are not promised in v0.3.0 (UI-090 is a Could for keyword highlighting).	M
FR-181	Run executes in a transaction marked READ ONLY with a statement_timeout from Settings (default 60 s). Statements that PostgreSQL rejects as writes return the server error unchanged plus a note 'DataWeave SQL is read-only in this version'.	M
FR-182	A text pre-check blocks obvious non-read statements (INSERT, UPDATE, DELETE, DDL, COPY ... FROM, CALL, DO, multiple statements in one run) with a clear explanation before sending. The pre-check is a convenience; FR-181 is the safeguard.	M
FR-183	Result grid reuses the Data grid component and its sort (client-side on loaded rows), selection, copy, NULL and type formatting. Results are fetched in pages of 500 using a server-side cursor; the grid indicates 'More rows available' and a Fetch more action. Sorting and filtering of results beyond loaded rows is not offered; the UI states this.	M
FR-184	A context chip shows connection and database; the query runs on that context. If the context changes while the editor has text, the text is kept and the chip updates. Running with no connection shows an inline Connect action, not a silent failure.	M
FR-185	Status: Idle, Running (elapsed timer), Done (rows, duration in ms), Error, Cancelled. Duration is measured from request to last fetched page.	M
FR-186	Errors show SQLSTATE, message, detail, hint and position when provided; the editor marks the line and column of the reported position.	M
FR-187	Cancel issues a JDBC cancel, then pg_cancel_backend if still running after 2 s; the connection returns to the pool in a clean state or is discarded.	M
FR-188	Clear empties the editor after confirmation if the text is unsaved and longer than 200 characters; separate Clear results.	M
FR-189	History: each run stores timestamp, connection id, database, SQL text, status, duration and row count in SQLite; retention default 1,000 entries or 90 days, configurable; Delete one, Clear all, search, Re-run, and Copy. Passwords and connection strings are never stored.	M
FR-190	Export results: CSV streamed by re-running the query with a cursor, with Cancel and progress, using FR-137 options.	M
FR-191	Multiple SQL tabs; each has its own text and results; tabs restore after restart (text only).	S
FR-192	Save query to a project from the editor toolbar (FR-203).	S
FR-193	EXPLAIN support: an Explain button runs EXPLAIN (not ANALYZE) and shows plan text.	C
UI-090	Basic keyword and string highlighting in the editor if achievable with reliable performance; otherwise omitted.	C

16. Projects

A project groups reusable database work. It stores references and settings in SQLite. It does not store PostgreSQL row data.

16.1 Contents

Item type	What is stored	Not stored
Dataset reference	Dataset id and display name	Result rows
Query	SQL text, name, target connection and database	Results
Table bookmark	Connection, database, schema, table; optional saved column selection, sort and filter	Rows
Export preset	Format, delimiter, header, NULL text, encoding, limit, last output folder	Exported files
Metadata	Name, description, created, modified, last opened	Passwords or tokens

16.2 Requirements

ID	Requirement	Priority
FR-200	Create project with name (unique, 1 to 80 characters) and optional description; the new project opens immediately.	M
FR-201	Open, Rename, Edit description, Duplicate (copies references, creating copies of datasets only if the user selects 'Also copy datasets'), Delete (confirmation naming item count; deleting a project never deletes datasets unless the user ticks that box).	M
FR-202	Add item from Datasets, SQL, Data (Add table to project) and from the project detail screen.	M
FR-203	Save is automatic on change with a visible Saved time; there is no state where edits exist only in memory without an indicator.	M
FR-204	Opening a project item opens it in the owning section (dataset in builder, query in a SQL tab, table in Data).	M
FR-205	Unavailable source: when the connection is not connected, the item shows 'Not connected' with Connect; when the profile was deleted, 'Connection removed' with Reassign connection; when the table or column is missing, 'Source missing' with details. Items remain in the project and can be repaired or removed.	M
FR-206	Export a project definition to a single JSON file and import it; the file contains no secrets and no row data; import maps connections by name with a chooser.	S
FR-207	Recent projects appear on Home; last open project is restored on startup if the setting is on.	S

17. Connections and Security

17.1 Connection lifecycle

Action	Behavior	Must not
Save	Stores profile metadata in SQLite and the password in the OS credential store; reports 'Saved'	Claim connected, or open a persistent session
Test	Opens a short-lived JDBC connection with the entered values, runs a trivial query, reports server version, latency and SSL status, then closes it	Change the active connection or show Connected in the status bar
Connect	Creates the pooled session, loads databases (FR-110), sets status Connected	Report Connected before the database list returns or fails
Disconnect	Closes pools, cancels running queries, clears metadata cache and shows Disconnected everywhere	Delete the profile or credential
Reconnect	Rebuilds the pool using the stored credential and restores Workspace Context	Loop silently; after 3 failures show Failed with Retry
Edit	Loads the selected profile into the form including non-secret fields; the password field shows a placeholder dots only when a credential exists	Display the stored password
Delete	Confirmation; removes profile and credential; closes its session	Leave orphan credentials

17.2 Requirements

ID	Requirement	Priority
FR-100	Profile form fields: display name, host, port (default 5432), database, username, password, SSL mode (disable, allow, prefer, require, verify-ca, verify-full), root certificate path when needed, connect timeout (default 10 s), application name. Validation is inline and prevents Save with empty host, invalid port or empty name.	M
FR-101	Profile list shows name, host, database and status. Duplicate and Delete are available.	M
FR-102	Save persists the profile without requiring a successful Test, and states that clearly; Save with password stores the password in the credential store (SEC-001).	M
FR-103	Test never alters the active session and shows result details (version, latency, SSL in use, or failure kind).	M
FR-104	Connect from Home, Connections, explorer and recovery panel use the same service and show the same statuses.	M
FR-105	Error kinds map to messages: server unreachable (host or port), authentication failed, database does not exist, SSL required or certificate failure, timeout, missing_credential, permission denied. Each message names the next action.	M
FR-106	Credential recovery: when the credential is missing the UI shows the recovery panel in the explorer, on Home and on Connections with masked password input, Save credentials, Test, Connect and Open Connections; on success the panel disappears and normal state resumes without restart.	M
FR-107	After an upgrade, existing profiles and credentials are preserved; a profile whose credential is absent shows the recovery panel rather than failing generically.	M
FR-108	Connection status (five values) is held by one state holder and drives the status bar, Home, explorer, Data, SQL and Datasets.	M
FR-021	Workspace Context (connection id, database, schema, selected object) is one shared state with a single writer (explorer and status-bar picker).	M

17.3 Security requirements

ID	Requirement	Priority
SEC-001	Passwords are stored only in the OS credential store (Windows Credential Manager with a persistent per-user mode; macOS Keychain; Linux Secret Service). They are never written to SQLite, project files, dataset definitions, exports, logs or release artifacts.	M
SEC-002	If a platform credential store is unavailable (for example a Linux system without Secret Service), the user is told and may choose Do not save password; there is no silent plaintext fallback.	M

ID	Requirement	Priority
SEC-003	Logs redact passwords, JDBC URLs containing credentials, and SQL parameter values by default. A 'verbose SQL logging' option logs statements only and shows a warning.	M
SEC-004	SSL modes verify-ca and verify-full validate certificates; failure shows the specific cause. The default for new profiles is prefer, with a visible indicator when the session is not encrypted.	M
SEC-005	Read-only enforcement for SQL, Data, Dataset preview and Export is by READ ONLY transactions and a restricted statement path; no UI path issues a write.	M
SEC-006	Identifiers in generated SQL are quoted; values are bound. Unit tests include identifier injection attempts (quotes, semicolons, comments).	M
SEC-007	Local files (~/.dataweave/) are created with per-user permissions; exported files are written only where the user chooses.	M
SEC-008	Release artifacts are scanned for secrets in CI (pattern scan for passwords and tokens) before upload; checksums are published.	S
SEC-009	Clipboard copies of grid data are plain data; copying a connection never includes the password.	M

18. Functional Requirements and Button Contracts

This section defines the mandatory control inventory. A control is complete only when its contract is implemented and its test in section 22 passes. Controls without a contract may not ship. When a feature is unavailable, the control is hidden or disabled with a tooltip stating the reason; no visible control may be a placeholder.

ID	Requirement	Priority
FR-300	Maintain a machine-readable control inventory file (controls.csv in docs) listing id, screen, label, contract fields below and the test id. A CI check fails when a registered control id has no contract row or when a Compose test tag has no inventory row.	M
FR-301	Every control with an asynchronous action has loading, success and failure states and prevents double activation while running.	M
FR-302	Every failure state offers Retry or a clear alternative, and keeps user-entered input.	M
FR-303	Disabled controls show a tooltip with the missing precondition.	M

18.1 Global and navigation controls

Control	Precondition	Operation and persistence	Success state	Failure and retry
Nav: Home, Connections, Data, SQL, Datasets, Projects	None	Switch section; last section stored in settings	Section shown, nav item selected	If a section fails to compose, show error panel with Reload section
Theme toggle	None	Cycle Dark, Light, System; write setting	Applies at once; persists after restart	If write fails, applies for session and shows warning
Settings	None	Open settings dialog; each change writes on commit	Value persists	Revert to previous value with message
Status bar: connection chip	Profile exists	Click opens connection menu: Disconnect, Reconnect, Switch profile	Menu shows current state	Not applicable
Status bar: database picker	Connected	Switch database; updates Workspace Context	Explorer and tabs follow	Error row; context unchanged
Panel splitters, explorer collapse	None	Resize or collapse; persist width	Layout restored on restart	Reset to default on corrupt value

18.2 Home and Connections controls

Control	Precondition	Operation and persistence	Success state	Failure and retry
Home: Connect (per profile)	Profile with credential	Connect (FR-104)	Status Connected; Open Data enabled	Error message with kind; Retry; if missing_credential show recovery panel
Home: New connection	None	Open Connections with blank form	Form focused on Name	Not applicable
Recovery: Save credentials	Password entered	Store in credential store; update profile fields	Inline 'Saved', Connect enabled	Message naming the credential store error; Retry
Recovery: Test	Required fields valid	FR-103 flow	Result line with version and latency	Error kind and next step
Recovery: Connect	Credential saved or entered	Save if needed then connect	Panel disappears; explorer loads	Error; panel remains
Recovery: Open Connections	Profile selected	Open Connections with that profile loaded	Correct profile in form	If profile deleted, message and list

Control	Precondition	Operation and persistence	Success state	Failure and retry
Form: Test / Save / Connect / Cancel	Valid form	Per section 17.1; Cancel discards unsaved edits after confirm	Distinct messages: Tested, Saved, Connected	Inline error; form data retained
Profile: Edit, Duplicate, Delete	Profile selected	Load, copy, or delete with confirmation	List updated and persisted	Message; list unchanged
SSL mode selector, Reveal password	None	Update form value; reveal auto-hides	Reflected in form	Not applicable

18.3 Explorer and Data controls

Control	Precondition	Operation and persistence	Success state	Failure and retry
Tree expander	Node has children	Lazy load children (FR-111)	Children listed	Error row with Retry
Explorer filter, clear	Connected	Filter loaded nodes (FR-113)	Matches shown	Searching state; Cancel
Explorer Refresh / Refresh all	Connected	Invalidate cache and reload	Updated tree, timestamp	Error row; previous tree kept
Context menu items	Node selected	Per FR-116	Action performed	Message; no partial action
Open data (double-click)	Table or view	Open Data tab and run first page	Rows shown	Banner with Retry
Data: Refresh	Tab open	Re-run query (FR-135)	Rows and time updated	Banner; rows retained dimmed
Data: Columns chooser	Tab open	Show, hide, reorder; persist per table	Grid updates	Not applicable
Data: Filter, Search, Clear filters	Tab open	Server-side (FR-125, FR-126)	Rows and count update	Banner; filter retained
Header click (sort)	Tab open	FR-124	Sort indicator and rows	Banner; previous sort restored
Pager: Previous, Next, Go to row, page size	Rows exist	FR-127; page size persisted	Rows and range update	Banner with Retry
Count exactly	Tab open	Run COUNT(*) with Cancel	Exact count replaces estimate	Timeout message; estimate stays
Export	Tab open	FR-137 dialog then streamed export	File saved; path and rows reported with Open folder	Error and partial file removed; Retry
Copy actions	Selection	FR-131	Toast 'Copied n cells'	Message if clipboard denied
Cancel	Query running	FR-136	Status Cancelled	If cancel fails, status Error with Reconnect
Show query	Tab open	FR-139	Popover with SQL	Not applicable

18.4 Dataset, SQL and Projects controls

Control	Precondition	Operation and persistence	Success state	Failure and retry
Datasets: New, Open, Duplicate, Delete	None, or selection	FR-174; writes SQLite	List updated	Message; list unchanged
Add table / drag column	Connected and table visible	UI-040 to UI-044	Column in list, card on canvas	Inline reason (UI-042)
Accept join, Use this, Add join	Two tables	FR-150 to FR-153	Join link shown, validation updates	Validation message
Join type selector	Join exists	FR-153	Link label updates; preview stale flag	Not applicable

Control	Precondition	Operation and persistence	Success state	Failure and retry
Alias edit	Column in list	FR-160, FR-161	Name updated	Inline validation
Filter builder, Sort builder	Dataset has columns	FR-162, FR-163	Stale preview indicator	Validation message
Preview, Refresh preview	Valid definition	FR-171, FR-172	Rows in preview grid	Error with Retry; definition unchanged
Save, Save as	Name present	Write definition JSON	Saved time, no dirty marker	Error; dirty marker stays
Export dataset	Valid definition	FR-175	File saved	Error; Retry
Open in SQL	Valid definition	FR-176	New SQL tab with text	Message
SQL: Run, Cancel, Clear, History, Export	Connected (Run)	FR-181 to FR-190	Status Done with duration	Error with position; Retry
Projects: New, Open, Rename, Duplicate, Delete, Add item, Remove item, Export, Import	Per action	FR-200 to FR-206	Autosaved, Saved time	Message; no partial changes

Complete screen-level inventories (every icon, tab, menu item, dialog action and tree expander) are produced in Phase 1 and Phase 2 as controls.csv following these contracts. TEST-070 verifies the file against the UI.

19. Architecture

The architecture is retained. The work is to enforce clear boundaries so features can be tested without the UI. Class and package names are UNVERIFIED and are mapped in the Phase 1 audit.

Layer	Responsibility	Rules
Compose Desktop UI	Screens, components, theming, drag-and-drop, keyboard handling	No JDBC, SQL or SQLite calls. Reads state, sends intents.
State holders / view models	One per screen plus shared WorkspaceContext and ConnectionState holders; expose immutable UI state as StateFlow	Run on the UI scope; delegate work to services on IO dispatchers; own cancellation of their jobs
Application services (Spring beans)	ConnectionService, MetadataService, TableDataService, QueryService, DatasetService, ProjectService, ExportService, SettingsService	Stateless where possible; return typed results or typed errors, never raw exceptions to UI
PostgreSQL session management	Pool per connection and database (HikariCP); health checks; borrow and release; cancel tokens	Max pool size small (default 4); read-only transaction wrapper for all reads; idle eviction
Metadata services	Catalog queries, cache (Caffeine), invalidation	Only catalog queries; TTL configurable
SQL execution	Statement execution, cursor fetch, timeouts, cancellation, error mapping (SQLSTATE to error kinds)	Single execution path used by Data, SQL, Dataset preview and Export
Dataset query generation	Pure Kotlin generator from definition to SQL plus parameters; validator	No I/O; exhaustively unit tested
SQLite persistence	Repositories over Flyway-managed schema	All schema changes by forward-only migrations; no raw DDL elsewhere
Credential storage	Interface with Windows, macOS and Linux implementations	Passwords only here; typed missing_credential result
Background operations	Coroutine scopes with structured concurrency; task registry showing running tasks in the status bar	Every task has cancel and timeout
Error reporting	Sealed AppError hierarchy mapped to messages, with technical detail and log correlation id	UI shows message and detail; logs hold the correlation id

19.1 Key flows

Flow	Sequence
Open table	Explorer intent to WorkspaceContext to TableDataViewModel to TableDataService (builds query with quoted identifiers and bound parameters) to QueryService (read-only transaction, timeout, cursor) to rows to grid state
Preview dataset	Builder state to DatasetDefinition to validator to generator to QueryService with preview limit to grid state; stale flag on definition change
Connect	ConnectionViewModel to ConnectionService to CredentialStore read (typed missing_credential) to pool creation to MetadataService database list to ConnectionState Connected or Failed
Cancel	Any running task id to registry to JDBC cancel then pg_cancel_backend via a separate pool connection

19.2 Requirements

ID	Requirement	Priority
FR-001A	startForGui() with headless(false) remains in the entry point and is covered by a smoke test and a packaged-app launch test on Windows.	M
NFR-010	No blocking JDBC or file I/O on the UI thread. A debug-mode watchdog logs UI frames over 100 ms.	M
NFR-011	All services are constructed through Spring and tested with injected fakes for CredentialStore and clock.	M
NFR-012	Architecture decisions that change the stack require an ADR in docs/adr with the evidence from Phase 1.	M

20. Data Model and Persistence

Local persistence uses SQLite under ~/.dataweave/ with Flyway. Table names below are proposed logical names. The Phase 1 audit reconciles them with existing tables; migrations must preserve existing profiles, history and datasets (DM-001).

Table	Key fields	Notes
connection_profile	id, name, host, port, database, username, ssl_mode, ssl_root_cert_path, connect_timeout_s, app_name, created_at, modified_at, last_connected_at	No password column. Credential key derived from id.
app_setting	key, value, modified_at	Theme, grid row height, page size, timeouts, last section, last project, retention
ui_layout	scope, key, value	Panel widths, column widths per table, open tabs
query_history	id, connection_id, database, sql_text, status, duration_ms, row_count, executed_at, error_kind	Retention applied on insert
saved_query	id, name, sql_text, connection_id, database, created_at, modified_at	Referenced by projects
dataset_definition	id, name, description, schema_version, definition_json, definition_hash, cached_sql, created_at, modified_at	definition_json is authoritative (DM-020)
project	id, name, description, created_at, modified_at, last_opened_at	
project_item	id, project_id, item_type, ref_id, display_name, position, settings_json	item_type: dataset, query, table_bookmark, export_preset
export_preset	id, name, format, options_json	No credentials, no paths containing secrets
flyway_schema_history	Managed by Flyway	Forward-only migrations

ID	Requirement	Priority
DM-001	Migrations are forward-only and tested by upgrading databases created by v0.2.4, v0.2.5 and v0.2.9 fixtures (to be generated from those releases) to v0.3.0 and asserting profiles, history and datasets remain readable.	M
DM-002	Before applying a migration the app copies the SQLite file to a timestamped backup in ~/.dataweave/backups (keep 3). If migration fails, the app starts in recovery mode with the backup path shown and does not delete data.	M
DM-020	Dataset definition JSON carries schemaVersion; the loader upgrades older versions in memory and saves the new version on next save; unknown future versions open read-only with a message.	M
DM-021	Deleting a profile does not delete datasets, queries or projects referencing it; they show Connection removed (FR-205).	M
DM-030	SQLite access is serialized through one writer; reads may be concurrent; all writes in transactions.	M
DM-040	No PostgreSQL row data is persisted locally except explicit user exports and the bounded in-memory grid window.	M

21. Non-Functional Requirements

Numeric values are targets pending baselines. NFR-001 defines how baselines are measured; results are recorded in docs/PERFORMANCE_BASELINE.md before targets are frozen. No result is claimed in this document.

ID	Requirement	Priority
NFR-001	Measurement method: run on a documented reference Windows machine (CPU, RAM, disk recorded), local PostgreSQL 16 or later with the generated benchmark database (tables of 1 thousand, 1 million and 20 million rows with mixed types), warm and cold app start, 10 runs per metric, report median and 95th percentile.	M
NFR-002	Targets: open a table and show the first page, median at most 1.0 s and p95 at most 2.5 s for a local database; scrolling the grid stays at 60 frames per second median on 1 million rows with no frame over 100 ms in p95 (loaded window).	M
NFR-003	Cold start to interactive Home at most 4 s median; warm at most 2 s. Measured from process start to first interactive frame.	S
NFR-004	Sort or filter on an indexed column of a 1 million row table returns the first page in at most 2 s p95 (server time dominates; the app overhead at most 150 ms).	S
NFR-005	Memory: grid window default 5,000 rows; browsing a 20 million row table must not grow heap beyond the window plus 200 MB over baseline; verified with a heap check during TEST-080.	M
NFR-006	CSV export of 1 million rows streams with constant memory (growth at most 100 MB) and reports progress at least once per second.	M
NFR-007	UI remains responsive during any query: input events processed within 100 ms p95 while a query runs.	M
NFR-008	Cancellation returns control to the UI within 2 s; timeouts are configurable: connect 10 s, metadata 15 s, query 60 s, preview 30 s.	M
NFR-009	Resource cleanup: Disconnect and app exit close all pools, statements and cursors; no leaked connections after 100 connect and disconnect cycles (pg_stat_activity check).	M
NFR-020	Startup: no network access is required; no telemetry; if the SQLite folder is not writable the app shows a clear error and offers an alternate folder.	M
NFR-021	Logging: rolling files under ~/.dataweave/logs (10 MB by 5 files); levels ERROR, WARN, INFO, DEBUG; correlation ids; redaction per SEC-003; Open log folder in Settings.	M
NFR-030	Upgrade compatibility: installing v0.3.0 over v0.2.x via MSI preserves profiles, credentials, history and settings (TEST-090).	M
NFR-040	Accessibility: all controls reachable by keyboard; visible focus; accessible names for icon buttons exposed to the Windows UI Automation tree where Compose supports it; text contrast at least 4.5:1 and non-text UI at least 3:1; no information conveyed by color alone; respects Windows text scaling up to 150 percent and display scaling up to 200 percent. Screen reader support level is verified with Narrator and documented honestly.	M
NFR-041	Theme persistence: selected theme applies on next launch before the first frame.	M
NFR-050	Localization: v0.3.0 is English only; strings are externalized for later translation.	S
NFR-051	Compatibility: PostgreSQL 12 through 18 are supported for catalog queries; tested on PostgreSQL 16 and 18 at minimum.	M

22. Testing and Acceptance Matrix

Verification rule. A feature is called verified only when its test was executed and the result recorded with date, build id and tester. Automated results come from CI or a local Gradle run. Manual Windows GUI results come from a human using the packaged installer. A green build is not acceptance. Statuses in the table below are all **Not executed** because this is a requirements document.

Level	Tooling	Notes
Unit	JUnit 5, Kotlin test; Compose UI test where reliable	Generators, validators, formatters, state holders, error mapping
Integration	Real PostgreSQL via Testcontainers or CI service on versions 16 and 18; SQLite temp files; fake CredentialStore	Metadata, data paging, cancellation, migrations
Regression	Matrix in docs/REGRESSION_MATRIX.md keyed by TEST ids and past defects	Run on every release candidate
Windows desktop (manual)	Packaged MSI on clean Windows 10 and 11 VMs and one physical laptop at 100, 125, 150 percent scaling	Scripted checklists with screenshots
Headless smoke	:desktop:runHeadlessSmoke	Startup path only; does not verify GUI

ID	Area	Test	Type	Status
TEST-001	Startup	App starts with startForGui and headless(false); no HeadlessException on Windows; packaged app launches	Auto + Manual	Not executed
TEST-010	Credentials	Save profile with password; restart app; credential still found	Auto (fake) + Manual	Not executed
TEST-011	Credentials	Delete credential externally; Connect returns missing_credential; recovery panel appears in explorer, Home and Connections	Auto + Manual	Not executed
TEST-012	Credentials	Edit connection from explorer opens the correct profile	Manual	Not executed
TEST-013	Connections	Test does not change active session; Save does not show connected	Auto	Not executed
TEST-014	Connections	Unreachable host, wrong password, wrong database, SSL failure map to the specified messages	Auto	Not executed
TEST-015	Connections	Disconnect and Reconnect restore context; 100 cycles leak no sessions	Auto	Not executed
TEST-016	Security	Passwords absent from logs, SQLite, exports, project JSON and installer files	Auto scan	Not executed
TEST-020	Explorer	Databases, schemas, tables, views, columns, keys and indexes match catalog queries on a seeded DB	Auto	Not executed
TEST-021	Explorer	Lazy load, timeout row and Retry, filter, refresh and cache invalidation	Auto + Manual	Not executed
TEST-022	Explorer	Role with limited privileges sees only permitted objects; empty states are correct	Auto	Not executed
TEST-030	Data grid	Row and column headers, sticky behavior, resize, hide and reorder, persistence	Manual + UI test	Not executed
TEST-031	Data grid	Sort single and multi-key; stable paging with tiebreaker	Auto	Not executed
TEST-032	Data grid	Filter operators per type, search, clear; injection strings are inert	Auto	Not executed
TEST-033	Data grid	Selection, keyboard navigation, copy variants, NULL and empty string distinction, long text, type formatting	Manual + UI test	Not executed
TEST-034	Data grid	Cancel a slow query; refresh preserves state; error and empty states	Auto + Manual	Not executed
TEST-035	Data grid	Read-only: no write statement is issued by any grid interaction (statement log assertion)	Auto	Not executed
TEST-036	Large data	20 million row table browse, jump, sort on indexed column, memory ceiling (NFR-005)	Auto + Manual	Not executed

ID	Area	Test	Type	Status
TEST-040	SQL	Run SELECT; duration and row count; errors with position; write statements rejected by session	Auto	Not executed
TEST-041	SQL	Cancel, history save, re-run, delete, retention, context switch	Auto + Manual	Not executed
TEST-042	Export	CSV export from Data, SQL and Dataset: delimiter, quoting, NULL, encoding, 1 million rows constant memory, cancel	Auto	Not executed
TEST-043	Import	CSV registration and preview remain working (existing README capability) and are labeled non-PostgreSQL sources	Auto + Manual	Not executed
TEST-050	Dataset	Generator unit tests: all four join types, composite keys, aliases, filters, sorts, self-join, identifier injection	Auto	Not executed
TEST-051	Dataset	FK discovery proposes joins; name-similarity hints never create joins automatically	Auto + Manual	Not executed
TEST-052	Dataset	Drag-and-drop to list and canvas, invalid drops, Escape cancel, keyboard alternatives, narrow-window layout	Manual + UI test	Not executed
TEST-053	Dataset	Save, close, reopen, modify; drift handling after dropping a column	Auto + Manual	Not executed
TEST-054	Dataset	Duplicate column names resolved before preview; unjoined table blocked	Auto	Not executed
TEST-055	Projects	Create, add items, reopen, duplicate, delete options, unavailable source states, export and import definition	Auto + Manual	Not executed
TEST-056	Theme	Theme persists across restart without flash; contrast checks pass in both themes	Auto + Manual	Not executed
TEST-060	Usability	Connect and open a table task with 10 first-time users	Manual	Not executed
TEST-061	Usability	Build a 3-table dataset and export task with 10 users	Manual	Not executed
TEST-070	Controls	controls.csv matches UI; every control exercised once on Windows; no dead controls	Auto + Manual	Not executed
TEST-080	Performance	Execute NFR-001 method; record baseline and compare to targets	Auto + Manual	Not executed
TEST-090	Upgrade	Install v0.2.5 and v0.2.9 MSIs, create profiles, history, datasets, then upgrade to v0.3.0; all preserved; migration failure recovery works	Manual + Auto	Not executed
TEST-091	Restart	Close and reopen with tabs, project and layout restored; sources missing handled	Manual	Not executed
TEST-092	Display	100, 125, 150, 200 percent scaling and two-monitor mixed DPI: no clipped controls, crisp text	Manual	Not executed
TEST-093	Packaging	MSI install, launch, uninstall on clean VM; checksum matches published file; Linux and macOS packages build	Manual + CI	Not executed

Repeated verification passes

- **Pass A (developer):** full automated suite plus targeted manual check on the developer machine after each phase.
- **Pass B (release candidate):** full automated suite on CI; packaged MSI on a clean Windows 11 VM; scripted manual checklist (TEST-001, 010 to 012, 030 to 035, 040, 042, 050 to 055, 056, 070, 092, 093).
- **Pass C (final):** second tester on Windows 10 physical laptop with the owner's real databases; upgrade test TEST-090; result sheet signed with build id.
- A release note may list a feature as Verified only when its TEST rows show an executed pass in Pass B or C.

23. Implementation Roadmap

Effort figures are rough planning ranges for a small team or coding agent and are not commitments. Phases 3 and 4 may overlap after the shared grid and state holders exist.

Phase 1: Repository audit and product baseline (about 1 week)

Field	Detail
Deliverables	Read all docs and source; produce current-state report; module and dependency map; list of Flyway migrations; initial controls.csv; reconcile README with release v0.2.9; generate benchmark database; record baselines (NFR-001); build upgrade fixtures; decide Compose grid approach with a spike (A-04).
Dependencies	Repository access and CI.
Risks	Docs and code diverge; hidden failures in Windows GUI.
Tests	Baseline report reviewed; TEST-080 baseline recorded; Windows manual pass of current build documented.
Deferred	No feature code except test fixtures.
Acceptance	Control inventory complete for existing screens; assumptions A-01 to A-07 marked confirmed or changed.

Phase 2: Design system and application shell (about 2 weeks)

Field	Detail
Deliverables	Tokens, typography, components, themes, nav rail, status bar, splitters, empty and error components, Settings dialog, high-DPI fixes.
Dependencies	Phase 1 findings on blur and fonts.
Risks	DPI blur needs JVM or Skiko flags; font licensing.
Tests	Screenshots at 4 scales; UI-001 to UI-032; TEST-056, TEST-092.
Deferred	Dataset builder visuals.
Acceptance	Every existing screen renders with new components; theme persists; no clipped controls at minimum window.

Phase 3: Connections and Explorer reliability (about 2 weeks)

Field	Detail
Deliverables	Single connection state holder, WorkspaceContext, five-state status, error kinds, recovery panel in all places, lazy explorer, cache, context menus, cancel and timeout infrastructure.
Dependencies	Phase 2 components.
Risks	Role permissions vary; stuck loading regressions.
Tests	TEST-001, 010 to 016, 020 to 022.
Deferred	Advanced explorer features such as drag from tree to canvas beyond list drag.
Acceptance	Connect to PostgreSQL 16 and 18 on Windows; databases, schemas, tables and columns load; every failure path has a message.

Phase 4: Spreadsheet Data workspace (about 3 weeks)

Field	Detail
Deliverables	Virtualized grid, TableDataService, sort, filter, search, selection, copy, formatting, paging, count, export, tabs, persistence.
Dependencies	Phase 3 session and cancellation.
Risks	Compose grid performance; huge tables; type edge cases.
Tests	TEST-030 to 036, 042, NFR-002 and NFR-005 measurement.
Deferred	Editing (v0.4.0).

Field	Detail
Acceptance	Open a 20 million row table and browse with memory ceiling met; all FR-120 to FR-142 pass.

Phase 5: Visual Dataset Builder (about 3 weeks)

Field	Detail
Deliverables	Definition model, generator, validator, canvas, drag-and-drop, FK discovery, joins, aliases, filter and sort, preview, save and reopen, export.
Dependencies	Phase 4 grid component; Phase 3 metadata.
Risks	Drag-and-drop complexity; join semantics; schema drift.
Tests	TEST-050 to 054.
Deferred	Cross-database and CSV joins; cross joins; aggregations.
Acceptance	WF-04 completes without SQL; saved dataset reopens identically; invalid drops handled; keyboard alternatives work.

Phase 6: SQL and Projects integration (about 2 weeks)

Field	Detail
Deliverables	SQL workspace on shared context, read-only enforcement, history, cancel, export; Projects with references and unavailable-source handling; Open in SQL; Save to project.
Dependencies	Phases 3 to 5.
Risks	Cancellation edge cases; project reference integrity.
Tests	TEST-040, 041, 055.
Deferred	Autocomplete, formatting, write mode.
Acceptance	WF-05 and WF-06 pass; project reopen restores items; a deleted connection is handled gracefully.

Phase 7: Security, persistence, performance, accessibility (about 2 weeks)

Field	Detail
Deliverables	Credential audit, log redaction, migration backups, accessibility pass, performance tuning against baseline, upgrade handling.
Dependencies	Features complete.
Risks	Accessibility gaps in Compose Desktop; migration edge cases.
Tests	TEST-016, 056, 080, 090, NFR-040.
Deferred	Full screen-reader parity if Compose cannot deliver it; documented honestly.
Acceptance	Security tests pass; targets met or exceptions approved by the owner with data.

Phase 8: End-to-end regression testing (about 1 week)

Field	Detail
Deliverables	Run the full matrix; fix defects; second tester; usability sessions.
Dependencies	Phase 7.
Risks	Late discovery of GUI defects.
Tests	TEST-060, 061, 070, 091 and all earlier tests re-run.
Deferred	Non-blocking polish.
Acceptance	Pass B complete with zero open severity 1 or 2 defects.

Phase 9: Windows packaging and release verification (about 1 week)

Field	Detail
Deliverables	MSI and EXE build, clean-VM install, upgrade test, checksums, release notes with an honest verification table, README correction.
Dependencies	Phase 8.
Risks	Installer or version-mapping issues; unsigned installer warnings.
Tests	TEST-090, 092, 093.
Deferred	Code signing improvements.
Acceptance	Pass C complete; release notes list executed tests only; README and releases agree.

24. Risks and Mitigations

ID	Risk	Likelihood / impact	Mitigation
R-01	Repeat of earlier defects where tests pass but the GUI fails	High / High	Windows manual passes in every phase; control inventory; Pass B and C rules; no Verified label without executed tests
R-02	Compose Desktop grid performance with large windows or wide tables	Medium / High	Phase 1 spike; virtualization; fall back to a canvas-drawn grid; measured against NFR-002
R-03	Blurry rendering at fractional Windows scaling	Medium / Medium	Vector assets only; Phase 2 DPI test matrix; JVM and Skiko settings if required
R-04	Scope creep into editing, DDL or cross-source joins	High / Medium	Section 8 exclusions; changes require owner approval and a roadmap update
R-05	Dataset definition drift as schemas change	Medium / Medium	FR-173 Needs attention state; recorded types; version field
R-06	Credential store behavior differs by OS or user policy	Medium / High	Typed errors; recovery UI; SEC-002; tests on clean VMs
R-07	Migration breaks existing profiles	Low / High	DM-001, DM-002 backups and fixtures from released versions
R-08	Long queries or huge tables exhaust memory or hold locks	Medium / High	Read-only transactions, timeouts, cursors, bounded window, cancel
R-09	Documentation drift between README, releases and installers	High / Low	Phase 9 reconciliation and a release checklist item
R-10	Limited accessibility support in Compose Desktop	Medium / Medium	Test with Narrator, document supported level, keyboard completeness as the baseline
R-11	Unknown repository facts invalidate assumptions	Medium / Medium	Phase 1 audit and assumption table updates before Phase 2 starts

25. Definition of Done

Feature level

- Requirement IDs implemented and linked to tests that were executed.
- All controls registered in controls.csv with contracts; no placeholder controls remain.
- Loading, empty, error and retry states exist and were seen in the running Windows app.
- Dark and light themes checked; narrow-window layout checked.
- No password, token or row data appears in logs, SQLite, exports or definitions.
- Docs updated: ARCHITECTURE.md, FEATURE_INVENTORY.md, VERIFICATION_STATUS.md, REGRESSION_MATRIX.md.

Release level (v0.3.0)

- All Must requirements pass; Should items either pass or have owner-approved deferral recorded.
- Pass A, B and C completed; zero open severity 1 and 2 defects; earlier defects (HeadlessException, scroll, loading stuck, missing credential, dead controls) have passing regression tests.
- Upgrade from v0.2.5 and v0.2.9 verified on Windows.
- Performance measured against NFR-001 and recorded; any missed target has a written exception.
- MSI and EXE verified on a clean Windows machine; checksums published; README, release notes and installers agree on the version.
- Release notes state which tests were executed and which areas remain unverified.

26. Final Implementation Checklist

Use this checklist as the work order. Mark an item only after its test has been executed.

Phase	Item
Phase 1	[ ] Read all repository docs and source
	[ ] Map architecture and migrations
	[ ] Create controls.csv for current screens
	[ ] Record NFR-001 baselines
	[ ] Create upgrade fixtures from v0.2.5 and v0.2.9
	[ ] Grid spike decision recorded
Phase 2	[ ] Tokens and fonts bundled
	[ ] Components with all states
	[ ] Dark, Light, System themes
	[ ] Nav rail, status bar, splitters
	[ ] High-DPI verified at 4 scales
	[ ] Settings dialog persists values
Phase 3	[ ] ConnectionState and WorkspaceContext single sources of truth
	[ ] Five status values consistent
	[ ] Credential recovery in explorer, Home, Connections
	[ ] Typed error kinds and messages
	[ ] Lazy explorer with cache and Retry
	[ ] Cancel and timeout infrastructure
Phase 4	[ ] Virtualized grid with sticky headers and row numbers
	[ ] Sort, filter, search, paging
	[ ] Selection, keyboard navigation, copy
	[ ] NULL, long text and type formatting
	[ ] Exact count on demand
	[ ] Streaming CSV export
	[ ] Tabs and persisted layout
	[ ] View-only guarantee tested
Phase 5	[ ] Definition model and versioned JSON
	[ ] SQL generator and validator
	[ ] Drag-and-drop with valid and invalid targets
	[ ] Keyboard alternatives
	[ ] FK joins and name-match hints
	[ ] Four join types
	[ ] Aliases, filters, sorts
	[ ] Preview, save, reopen, drift handling, export
Phase 6	[ ] SQL read-only enforcement by transaction
	[ ] History, cancel, errors with position
	[ ] Result export
	[ ] Projects CRUD and references
	[ ] Unavailable-source handling

Phase	Item
	[ ] Open in SQL and Save to project
Phase 7	[ ] Credential, log and artifact secret scans
	[ ] Migration backup and recovery
	[ ] Accessibility checks
	[ ] Performance against targets
	[ ] Theme persistence
Phase 8 and 9	[ ] Full regression matrix executed
	[ ] Usability sessions
	[ ] MSI and EXE clean-VM verification
	[ ] Upgrade test
	[ ] Release notes with executed-test table
	[ ] README and releases reconciled